

Data Dictionary: Hospital Emergency Room (ER) Operations

Project Context: Emergency Department Patient Flow and Experience Analysis

Source Data: Operational records covering patient demographics, admission outcomes, and satisfaction metrics.

Feature Name	Data Type	Description	Operational Significance
Patient Id	String	A unique, hyphenated identifier for each patient.	Used for record tracking and preventing data duplication.
Patient Admission Date	DateTime	Timestamp of patient arrival at the ER.	Critical for identifying peak arrival times and seasonal volume shifts.
Patient First Initial	String	The first letter of the patient's legal first name.	Used for basic identity verification and privacy-safe reporting.
Patient Last Name	String	The legal surname of the patient.	Used for patient identification and record matching.
Patient Gender	Categorical	Patient gender: Male (M), Female (F), or Non-conforming (NC).	Essential for demographic reporting and clinical resource planning.

Patient Age	Integer	The age of the patient (Range: 1 to 79).	Used to segment care needs (e.g., pediatric vs. geriatric priorities).
Patient Race	Categorical	The self-identified race of the patient.	Used for analyzing equity in healthcare delivery and access.
Department Referral	Categorical	The specialist department assigned to the patient.	Tracks demand across Cardiology, Neurology, Orthopedics, and others.
Patient Admission Flag	Boolean	Indicates if the patient was admitted to the hospital (True/False).	A primary outcome metric for ER throughput and bed capacity planning.
Patient Satisfaction Score	Float	Patient feedback score (Range: 1 to 10).	A key performance indicator (KPI) for service quality and patient experience.
Patient Waittime	Integer	Time in minutes from arrival to clinical assessment (Range: 10 to 60).	The critical efficiency metric for ER operational performance.
Patients CM	Boolean	Case Management flag (1 = Case Managed, 0 = Standard).	Identifies patients requiring high-touch coordination of care.

Data Observations and Governance

- Missing Value Management:** The **Patient Satisfaction Score** column contains a high volume of missing values (approximately 72% missing). This suggests that

satisfaction surveys are optional or only captured for specific patient cohorts, requiring careful handling in predictive modelling.

- **Referral Logic:** The Department Referral field is null for patients who were likely discharged without specialist follow-up or whose cases were resolved within General Practice.
- **Operational Scope:** The dataset covers 9,216 patient records, providing a robust statistical sample for analysing wait-time drivers and admission probabilities.
- **Categorical Diversity:** The Patient Race and Patient Gender fields include "Declined to Identify" and "NC" categories, respectively, ensuring inclusive data collection standards.